

Bocconi University

20592 Statistics and Probability

Final Project

Group 17

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Introduction & datasets

In this project, generalized linear models (GLMs) are examined using two different datasets. The aim is to show how an appropriate statistical model can be constructed and how parameter estimation can be performed for various problems involving different types of response variables. In the CreditCard dataset, the response variable is binary, whereas in the second dataset, azcabgptca, the response variable is a count variable. Logistic regression is considered for the first dataset, while Poisson regression is used for the second one.

1. a) Fisher Scoring Implementation

For each dataset, parameter estimation is performed using the Fisher scoring algorithm, implemented from scratch. Let X denote the design matrix and y the response variable. At iteration t , the linear predictor is defined as $\eta^{(t)} = X\beta^{(t)}$. Here, β denotes the vector of unknown regression coefficients associated with the covariates in the model.

For the logistic regression model, the conditional mean is given by $\mu(t) = \frac{1}{1 + \exp(-\eta^{(t)})}$,

while for the Poisson regression model with canonical log link, the conditional mean is given by $\mu^{(t)} = \exp(\eta^{(t)})$.

In both cases, the score function is defined as $s(\beta^{(t)}) = X^T(y - \mu^{(t)})$ and the Fisher information matrix is $I(\beta^{(t)}) = X^T W^{(t)} X$, where the weight matrix $W^{(t)}$ is given by $W^{(t)} = \text{diag}(\mu^{(t)}(1 - \mu^{(t)}))$ for the logistic model and $W^{(t)} = \text{diag}(\mu^{(t)})$ for the Poisson model. The Fisher scoring update is therefore $B^{(t+1)} = B^{(t)} + I(B^{(t)})^{-1} s(B^{(t)})$.

The algorithm is run iteratively until convergence, which is determined by the size of the update step. During the iterations, a step-halving procedure is applied to ensure that the log-likelihood does not decrease. After convergence, standard errors are obtained from the inverse of the Fisher information matrix evaluated at the final parameter estimates. This approach is well suited for generalized linear models, as it directly uses the score function and the Fisher information.

1. b) Fisher Scoring vs Newton

In a GLM, parameter estimation is based on maximizing the log-likelihood $l(\beta)$. Newton's method updates the parameter vector according to $\beta^{(t+1)} = \beta^{(t)} - [\nabla^2 l(\beta^{(t)})]^{-1} \nabla l(\beta^{(t)})$ where $\nabla l(\beta)$ is the score function and $\nabla^2 l(\beta)$ is the Hessian matrix.

Fisher Scoring replaces the observed Hessian $-\nabla^2 l(\beta)$ with its expectation, the Fisher information matrix $I(\beta)$, leading to an update $B^{(t+1)} = B^{(t)} + I(B^{(t)})^{-1} s(B^{(t)})$ where $s(\beta) = \nabla l(\beta)$.

In a GLM, the log-likelihood can be written in terms of μ_i and the linear prediction $\eta_i = x_i^T \beta$.

Taking derivatives w.r.t. β , the score function is $\nabla l(\beta) = \sum_{i=1}^n \frac{(y_i - \mu_i)}{\text{Var}(y_i)} \frac{\partial \mu_i}{\partial \eta_i} x_i$.

Differentiating once more gives the Hessian

$$\nabla^2 l(\beta) = -X^T W X + \sum_{i=1}^n (y_i - \mu_i) \frac{\partial}{\partial \beta} \left(\frac{1}{\text{Var}(y_i)} \frac{\partial \mu_i}{\partial \eta_i} \right) x_i^T, \text{ which in general depends on } y.$$

For models with a canonical link function, the natural parameter equals the linear predictor, $\eta_i = x_i^T \beta$. In this case $\frac{\partial \mu_i}{\partial \eta_i} = \text{Var}(y_i)$, so the term involving $(y_i - \mu_i)$ disappears.

As a result, the second derivative of the log-likelihood does not depend on the response variable y , but only on the current parameter value through mean μ . Therefore, the observed information matrix equals its expectation $-\nabla^2 l(\beta) = I(\beta)$.

Since the observed Hessian and the Fisher information matrix coincide for canonical link functions, Newton's method and Fisher scoring produce identical update steps. Consequently, the two algorithms are equivalent in this setting.

2. Markov Chain Monte Carlo Method

Bayesian inference is adopted for the generalized linear models by placing prior distributions on the regression coefficients and approximating the resulting posterior distributions using a Markov Chain Monte Carlo (MCMC) method. We consider a Bayesian logistic regression for the CreditCard dataset and a Bayesian Poisson regression for the azcabgptca dataset. Since the posterior distributions do not admit closed-form expressions, we implement a random-walk Metropolis–Hastings algorithm to generate samples from the posterior. To improve sampling efficiency, we scale the random-walk proposal using the Fisher information matrix obtained from Task 1.

Prior and posterior

Building on the likelihood formulations introduced in Task 1, we specify prior distributions on the regression coefficients to complete the Bayesian model. For both models, we assign independent Gaussian priors to the regression coefficients: $\beta_j \sim N(0, \sigma^2)$, $j = 1, \dots, p$, with

prior variance $\sigma^2 = 100$. These priors are weakly informative and provide mild regularization, while allowing the likelihood to dominate posterior inference.

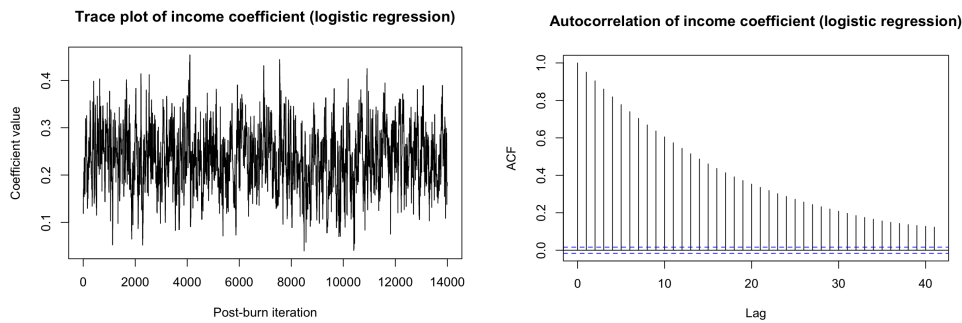
The log-prior is $\log p(\beta) = -\frac{1}{2\sigma^2}\beta^\top \beta + C$. The log-posterior distribution is therefore given by $\log p(\beta|y) = \ell(\beta) - \frac{1}{2\sigma^2}\beta^\top \beta + C$. This choice discourages unrealistically large coefficient values and improves numerical stability of the MCMC algorithm without imposing strong prior beliefs on the parameters.

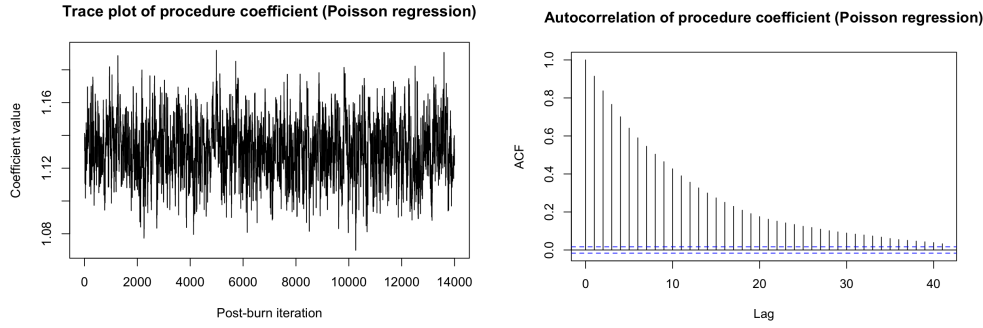
Metropolis–Hastings algorithm

Given the current state β^t , a proposal is generated as $\beta^{(*)} = \beta^t + sI(\hat{\beta})^{-1/2}z$, $z \sim N(0, I)$, where $I(\hat{\beta})$ denotes the Fisher information matrix evaluated at the Fisher scoring estimate $\hat{\beta}$. This proposal accounts for the local curvature of the posterior distribution and adapts the proposal scale to the correlation structure of the regression coefficients. Since the proposal distribution is symmetric, the acceptance probability simplifies to $\alpha = \min(1, \exp[\log p(\beta^{(*)}|y) - \log p(\beta^t|y)])$.

Convergence diagnostics

Each chain was run for 20,000 iterations, with the first 6,000 iterations discarded as burn-in. Proposal variances were tuned to obtain reasonable acceptance rates. To assess convergence and mixing, we examine trace plots and autocorrelation functions (ACF) for the coefficient associated with income in the logistic regression and procedure in the Poisson regression. The trace plots indicate that, after burn-in, the chains fluctuate within stable regions of the parameter space without systematic trends, suggesting stationarity. The ACF plots decay steadily with increasing lag, indicating substantially reduced autocorrelation and improved mixing. This improvement is due to scaling the random-walk proposal using the Fisher information matrix, which aligns the proposal distribution with the local geometry of the posterior.





Interpretation of the fitted models

CreditCard dataset

The variables share and expenditure are excluded because they are only meaningful after a credit card has been granted and used. As such, they do not represent information available at the time of application and would introduce post-decision information and target leakage. Therefore, we restrict the model to pre-application covariates.

For the credit card acceptance model, the posterior results indicate that “reports” ($\beta \approx -1.75$, $z \approx -29.5$), “income” ($\beta \approx 0.23$, $z \approx 4.21$) and “active” ($\beta \approx 0.13$, $z \approx 9.54$) are among the three most influential covariates. The number of credit reports has a strong negative effect, with an estimated odds ratio of $\exp(-1.75) \approx 0.17$, indicating that each additional reported credit issue reduces the likelihood of acceptance by approximately 83%. Income has a positive effect ($\exp(0.23) \approx 1.26$), suggesting that higher income increases the chance of approval by approximately 26%. The number of active accounts also has a positive effect ($\exp(0.13) \approx 1.14$), implying that greater engagement with the credit system is associated with a higher probability of acceptance.

azcabgptca dataset

For the hospital length-of-stay model, three most influential covariates are “procedure” ($\beta \approx 1.13$, $z \approx 66.6$), “type” ($\beta \approx 0.19$, $z \approx 11.2$) and “died” ($\beta \approx -0.31$, $z \approx -5.4$). The estimated coefficient for procedure ($\exp(1.13) \approx 3.10$) implies that CABG patients have an expected length of stay more than three times that of PTCA patients. Treatment type increases expected length of stay by approximately 21% ($\exp(0.19) \approx 1.21$), likely reflecting differences in treatment complexity. Patients who died tend to have shorter observed stays due to early termination of hospitalization and have an expected length of stay about 26% shorter ($\exp(-0.31) \approx 0.74$).

Non-canonical link discussion

In the previous sections, we showed that for GLMs with a canonical link function, the score and curvature of the log-likelihood take a particularly simple form, implying that the observed

Hessian coincides with the Fisher information matrix and that Fisher scoring and Newton's method produce identical update steps.

When a non-canonical link function is used, the relationship between the linear predictor and the mean response becomes $\eta = g(\mu)$, where g is not the canonical link. In this case, the score function can be written as $s(\beta) = X^T W(y - \mu)$

where the weight matrix now involves the derivative of the inverse link function. Moreover, the second derivative of the log-likelihood contains additional terms depending on y , so that $-\nabla^2 l \neq I(\beta)$

Fisher scoring therefore replaces the observed Hessian with its expectation, leading to update steps that differ from those of Newton's method.

A classical example is probit regression for binary data, where the link function is given by

$g(\mu) = \Phi^{-1}(\mu)$, with Φ denoting the standard normal cumulative distribution function. Since this link is not canonical for the Bernoulli distribution, the simplifications observed in logistic regression no longer apply. Nevertheless, Fisher scoring remains applicable by using the expected information matrix, although the algorithm may converge more slowly than in the canonical case.